

2024 AMC 10B Problem 15 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 15. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 15

**Problem:**

A list of 9 real numbers consists of 1, 2.2, 3.2, 5.2, 6.2, and 7, as well as x , y , and z with $x \leq y \leq z$. The range of the list is 7, and the mean and the median are both positive integers. How many ordered triples (x, y, z) are possible?

Answer Choices:

- A. 1
- B. 2
- C. 3
- D. 4
- E. infinitely many

Solution:

Because the range is 7, the values of x , y , and z are in the interval $[0, 8]$. Because the median is an integer, it is one of x , y , or z and is either 3, 4, 5, or 6. The sum of the list is $s = 24.8 + x + y + z$, which is between $24.8 + 0 + 0 + 3 = 27.8$ and $24.8 + 6 + 8 + 8 = 46.8$. Because the mean is an integer, s is an integer multiple of 9, so $s = 36$ or 45 , and $x + y + z = 11.2$ or 20.2 .

- ▶ If the median is 3, then $x \leq y \leq z = 3$, so $x + y + z \leq 9 < 11.2$. Therefore this case cannot occur.
- ▶ If the median is 4, then $x \leq y = 4 \leq z$, so $x + z = 7.2$, and the list spans one of the intervals $[0, 7]$, $[1, 8]$, or $[x, z]$. If $x = 0$, then $z = 7.2 > 7$, and if $z = 8$, then $x = -0.8 < 0$. Therefore these cases cannot occur. Otherwise $z - x = 7$, giving $(x, y, z) = (0.1, 4, 7.1)$.

- If the median is 5, then $x \leq y = 5 \leq z$, so $x + z = 6.2$. In order to have a range of 7, x must be 0, and $(x, y, z) = (0, 5, 6.2)$.
- If the median is 6, then $x = 6 \leq y \leq z$, so $y + z = 14.2$. In order to have a range of 7, z must be 8, and $(x, y, z) = (6, 6.2, 8)$.

Thus there are (C) 3 possible ordered triples (x, y, z) .

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